

Basics of Computer Networking

A computer network is a collection of interconnected devices that share resources and information. These devices include computers, servers, printers, and other hardware. Networks allow the efficient exchange of data, enabling applications such as email, file sharing, and Internet browsing.

Basic Terminology of Computer Networks

Network: A group of connected computers and devices that can communicate and share data.

Node: Any device that can send, receive, or forward data in a network. Examples include laptops, mobile phones, printers, earbuds, and servers.

Networking Devices: Devices that manage and support networking functions, such as routers, switches, hubs, and access points.

Transmission Media: The physical or wireless medium through which data travels.

- **Wired media:** Ethernet cables, optical fiber.
- **Wireless media:** Wi-Fi, Bluetooth, infrared.

Service Provider Networks: Networks provided by external organizations that offer communication services, such as Internet Service Providers (ISPs) and mobile network operators.

How Does a Computer Network Work?

The basic building blocks of a computer network are **nodes** and **links**.

Network Node: Any device in a network, such as a computer, modem, or router, that sends or receives data.

Link: The medium that connects nodes, including cables or wireless signals.

Working Principle: Networks use rules called **protocols** to exchange data. Each device is identified by a unique **IP address**.

Firewall: A hardware or software security system that monitors network traffic and allows or blocks data according to predefined rules.

Types of Computer Network Architecture

Client–Server Architecture:

In this architecture, nodes are divided into servers and clients. Servers manage resources and services, while clients request these services.

Peer-to-Peer (P2P) Architecture:

There is no central server. Each device can act as both a client and a server, sharing resources directly.

Network Devices

Computer networks rely on various devices that facilitate communication.

Common Network Devices

Router

- Connects multiple networks.
- Directs data packets to their destination.

Switch

- Connects devices within a local network.
- Sends data only to the intended device.

Hub

- Connects multiple devices.
- Broadcasts data to all devices.

Bridge

- Connects network segments.
- Reduces network traffic.

Gateway

- Connects different types of networks.
- Translates protocols.

Access Point (AP)

- Provides wireless access.

- Extends wired networks.

Modem

- Converts digital data into transmission signals.
- Provides Internet access.

Firewall

- Controls incoming and outgoing traffic.
- Protects against unauthorized access.

Goals of Computer Networks

- Convenience
- Efficiency
- Resource management
- Security and protection
- Reliability and fault tolerance
- Scalability

Uses of Computer Networks

- Communication
- Resource sharing
- Remote access
- Collaboration
- E-commerce
- Education

Characteristics of Computer Networks

Security: Protects data using firewalls, encryption, and authentication.

Reliability: Ensures continuous availability through backups and redundancy.

Scalability: Supports growth without performance degradation.

High Performance: Depends on bandwidth, latency, and throughput.

Quality of Service (QoS): Prioritizes critical data such as voice and video.

Technical English – Key Vocabulary

English Term	French Meaning
Network	Réseau
Node	Nœud (appareil du réseau)
Link	Lien (support de communication)
Networking Devices	Équipements réseau
Transmission Media	Support de transmission
Service Provider Network	Réseau d'un fournisseur de services
Router	Routeur
Switch	Commutateur
Hub	Concentrateur
Bridge	Pont
Gateway	Passerelle
Access Point	Point d'accès
Modem	Modem
Firewall	Pare-feu
Architecture	Architecture
Client–Server	Client-serveur
Peer-to-Peer (P2P)	Pair-à-pair
Bandwidth	Bande passante
Reliability	Fiabilité
Scalability	Scalabilité
Quality of Service (QoS)	Qualité de service